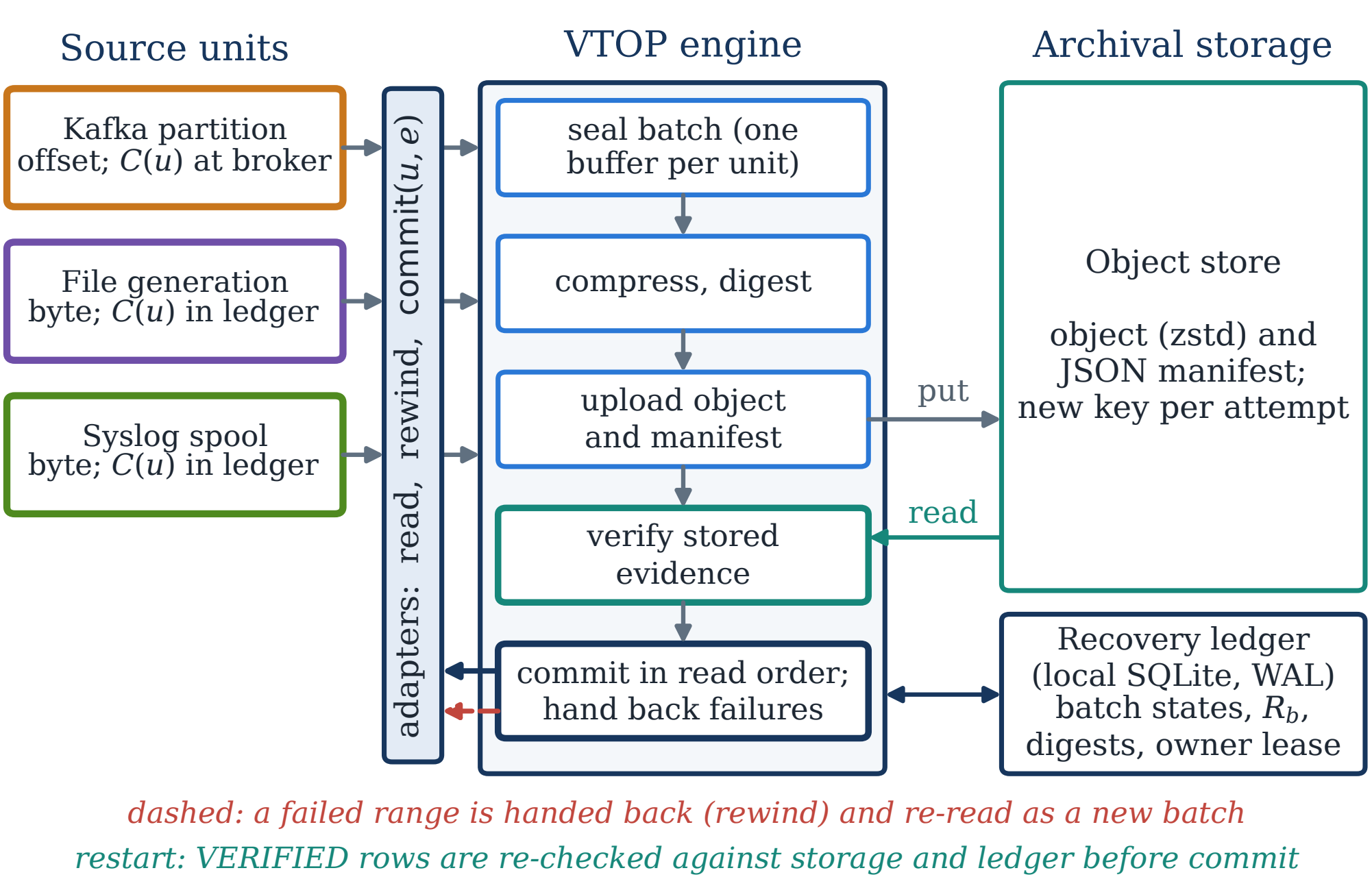
Source units
VTOP engine
Archival storage
Kafka partition offset; $C(u)$ at broker
File generation byte; $C(u)$ in ledger
Syslog spool byte; $C(u)$ in ledger
adapters: read, rewind, commit(u, e)
seal batch (one buffer per unit)
compress, digest
upload object and manifest
verify stored evidence
commit in read order; hand back failures
put
read
Object store
object (zstd) and JSON manifest; new key per attempt
Recovery ledger (local SQLite, WAL) batch states, R_b, digests, owner lease
dashed: a failed range is handed back (rewind) and re-read as a new batch
restart: VERIFIED rows are re-checked against storage and ledger before commit